From Neighbors to Probabilities: ML Models for Breast Cancer Detection

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Introduction

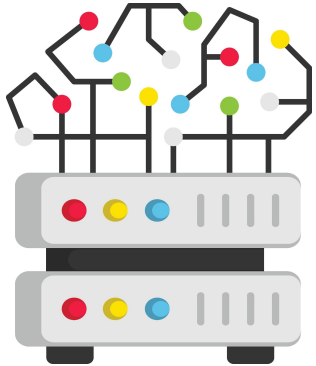
- Breast cancer is one of the most common cancers affecting women in the United States, accounting for approximately 30% of new cancer diagnoses among women each year
- Machine Learning (ML) offers powerful tools to support early detection and diagnostic decision-making by analyzing clinical and biological data to improve classification accuracy
- Two supervised learning algorithms were implemented: Logistic Regression and K-Nearest Neighbors (KNN) will be implemented to evaluate the performance of the model

Problem Statement

- Breast cancer is the most common form of cancer globally, accounting for 1 in 8 new cancer diagnoses worldwide
- When breast cancer is identified at stage 1, more than 99% of women survive for at least five years, highlighting the critical, lifesaving role of early detection.
- The average per patient cost of medical services during the initial care phase for breast cancer was $35,000, these cost double during the end-of-life phase at $76,100

Purpose of the Code

- Statistically, the diagnosis and treatment of breast cancer continues to be a significant burden to patients and healthcare systems alike.
- Given the variability in diagnostic accuracy associated with current breast cancer diagnostic tools, machine learning (ML) techniques offer a promising solution



Code Background

- The [Breast Cancer Wisconsin Diagnostic Dataset from the UCI Machine Learning Repository]() is a well-known dataset that will be utilized to conduct the ML model training and validation sets
- The dataset facilitates the analysis of 569 instances and 30 numerical predictive properties, including classes, to facilitate the diagnosis of cancerous breast tissue
- The nine predictive attributes that the dataset is structured around are rated on a numerical scale from 1 to 10

```
=== DATASET SHAPE ===
(569, 30)

=== CLASS DISTRIBUTION ===
1    357
0    212
Name: count, dtype: int64

=== FEATURE NAMES ===
['mean radius' 'mean texture' 'mean perimeter' 'mean area'
 'mean smoothness' 'mean compactness' 'mean concavity'
 'mean concave points' 'mean symmetry' 'mean fractal dimension'
 'radius error' 'texture error' 'perimeter error' 'area error'
 'smoothness error' 'compactness error' 'concavity error'
 'concave points error' 'symmetry error' 'fractal dimension error'
 'worst radius' 'worst texture' 'worst perimeter' 'worst area'
 'worst smoothness' 'worst compactness' 'worst concavity'
 'worst concave points' 'worst symmetry' 'worst fractal dimension']

=== TARGET CLASSES ===
['malignant' 'benign']

=== FEATURE DESCRIPTIONS (WITH UNITS) ===
mean radius: mm (tumor radius)
mean texture: std of gray-scale values
mean perimeter: mm
mean area: mm^2
mean smoothness: unitless
mean compactness: unitless
mean concavity: unitless
mean concave points: unitless
mean symmetry: unitless
mean fractal dimension: unitless

=== SUMMARY STATISTICS ===
        mean radius  mean texture  mean perimeter    mean area  \
count   569.000000    569.000000      569.000000   569.000000
mean     14.127292     19.289649       91.969033   654.889104
std       3.524049      4.301036       24.298981   351.914129
min       6.981000      9.710000       43.790000   143.500000
25%      11.700000     16.170000       75.170000   420.300000
50%      13.370000     18.840000       86.240000   551.100000
75%      15.780000     21.800000      104.100000   782.700000
max      28.110000     39.280000      188.500000  2501.000000
```

Train-Test Split and Feature Scaling

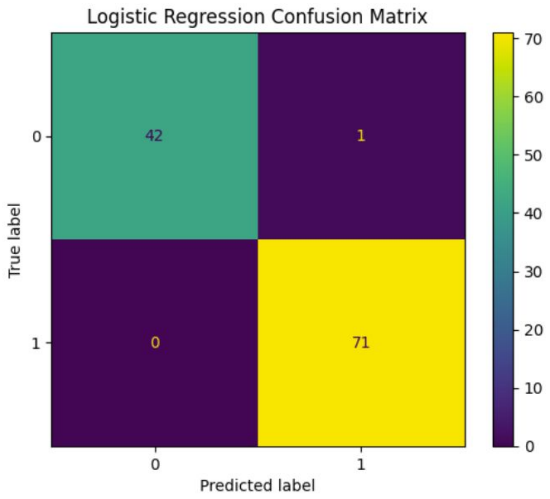
```python
# ========================
# TRAIN TEST SPLIT
# ========================
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```python
# ========================
# FEATURE SCALING
# ========================
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Logistic Regression

- Logistic regression was chosen as an algorithm to the train and test the model because the dataset is largely linear separable based on tumor features such as radius, area, and texture.

```
=== LOGISTIC REGRESSION ===
Accuracy: 0.9912280701754386
Precision: 0.9861111111111112
Recall: 1.0
F1 Score: 0.993006993006993

Classification Report:

              precision    recall  f1-score   support

           0       1.00      0.98      0.99        43
           1       0.99      1.00      0.99        71

    accuracy                           0.99       114
   macro avg       0.99      0.99      0.99       114
weighted avg       0.99      0.99      0.99       114
```

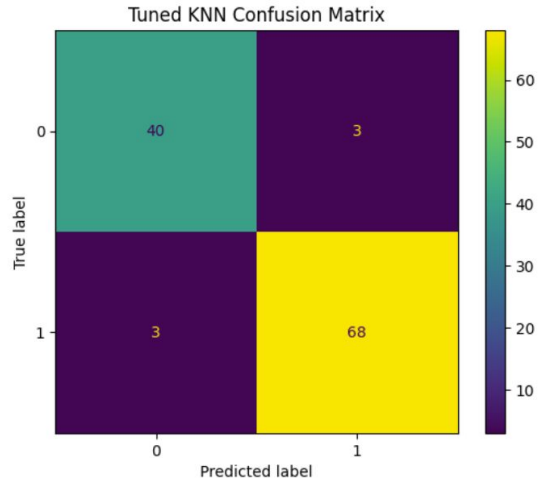



```
Logistic Regression False Negatives: 0
→ 0 malignant tumors were misclassified as benign.
```

K-Nearest Neighbor

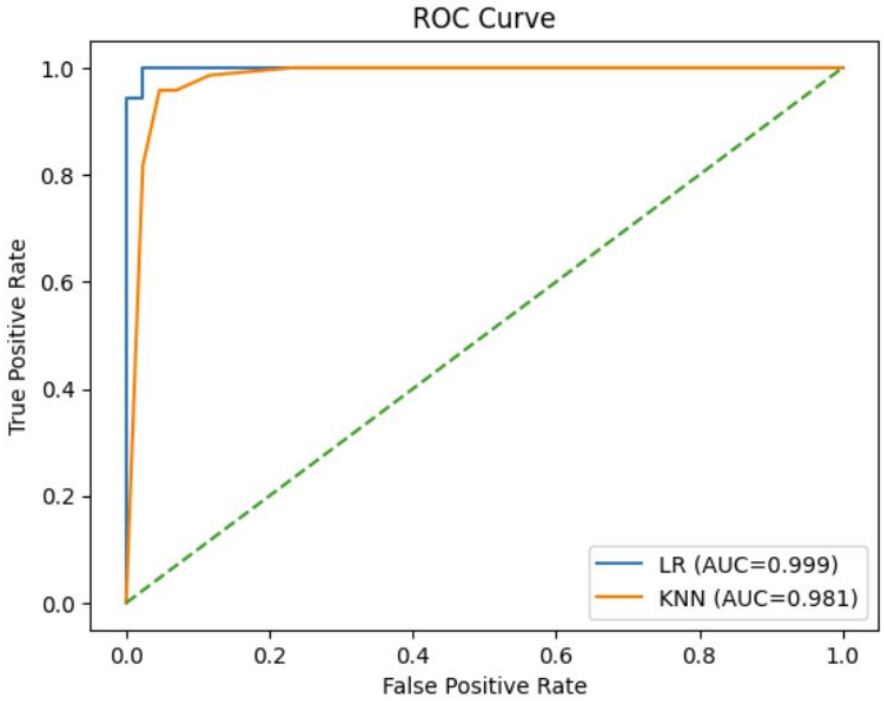
- KNN was chosen because the features, such as radius, area, and concavity, included in the dataset tend to form distinct clusters

```
=== TUNED KNN ===
Accuracy: 0.9473684210526315
Precision: 0.9577464788732394
Recall: 0.9577464788732394
F1 Score: 0.9577464788732394

Classification Report:

              precision    recall  f1-score   support

           0       0.93      0.93      0.93        43
           1       0.96      0.96      0.96        71

    accuracy                           0.95       114
   macro avg       0.94      0.94      0.94       114
weighted avg       0.95      0.95      0.95       114
```




```
KNN False Negatives: 3
→ 3 malignant tumors were misclassified as benign.
→ In a clinical setting, these represent missed cancer diagnoses.
```

ROC Curve and AUC scores for Logistic Regression and KNN




```
=== AUC SCORES ===
Logistic Regression: 0.998689813298395
KNN: 0.980510972813626
```

Conclusion

- In an article published by Mohammed Amine Naji et al. called *Machine Learning Algorithms for Breast Cancer Prediction*, the Breast Cancer Wisconsin dataset was used to train a Machine Learning (ML) model using similar algorithms to those used to train a this ML model
  - Logistic regression
    - Accuracy measure - 95.8%
    - F1-score - 0.94
    - AUC Score - 0.947
  - KNN
    - Accuracy measure - 93.7%
    - F1-score - 0.91
    - AUC Score - 0.952